

INDEX

Material Order Module – Overview

MaterialOrder Entity

MaterialOrderItem Entity

MaterialOrderService – Overview

MaterialOrderService – Core Methods

MaterialOrderController – Overview

Helper –
CheckLoggedInUserIfTrainerAndReturnData

Controller Actions – Main

Controller Actions – Details & Reporting

OTP & SMS Approval Workflow

Material Order Module – Full Documentation

Material Order Module – Overview

Purpose	Manage material orders for trainers and managers with filtering, printing, Excel export, and OTP-based approval.
Main Components	MaterialOrderController – MVC Controller (Admin Area) MaterialOrderService – Application Service MaterialOrder & MaterialOrderItem – Domain Entities
Key Features	Create and edit material orders Filter and view lists by user, department, and date range Print list and single orders Export filtered list to Excel (AR/EN) OTP-based Trainer / Manager signatures

INDEX

Material Order Module – Overview
MaterialOrder Entity
MaterialOrderItem Entity
MaterialOrderService – Overview
MaterialOrderService – Core Methods
MaterialOrderController – Overview
Helper – CheckLoggedInUserIfTrainerAndReturnData
Controller Actions – Main
Controller Actions – Details & Reporting
OTP & SMS Approval Workflow

MaterialOrder Entity

Id	Primary Key (int)
MaterialOrderCode	Required string, max length 50 Sequential code generated by service (GetNewCodeAsync)
DepartmentId	Required int? Foreign key to Department
Department	Navigation property to Department
UserId	Required string? Application user who created the order
TrainerId	Optional trainer reference (int?) for logged trainer scenario
Date	Required DateOnly ? Stored as date column (order date)
OrderTotal	Decimal? (default 0.00) Sum of (Quantity × SinglePrice) for all items Calculated and maintained in service
Items	ICollection<MaterialOrderItem> Collection of order line items
Notes	Optional notes (string?, max length 500)
SignatureUserId	int? Foreign key to Signature for Trainer signature
SignatureUser	Navigation property to trainer Signature
SignatureManagerId	int? Foreign key to Signature for Manager signature
SignatureManager	Navigation property to manager Signature

INDEX

Material Order Module – Overview
MaterialOrder Entity
MaterialOrderItem Entity
MaterialOrderService – Overview
MaterialOrderService – Core Methods
MaterialOrderController – Overview
Helper – CheckLoggedUserIfTrainerAndReturnData
Controller Actions – Main
Controller Actions – Details & Reporting
OTP & SMS Approval Workflow

MaterialOrderItem Entity

MaterialOrderItemId	Primary Key (int)
MaterialOrderId	Required int Foreign key to MaterialOrder
MaterialOrder	Navigation property to MaterialOrder
Quantity	int?, default 0 Optional, validated as numeric
SinglePrice	decimal?, default 0.00 Range validation to ensure positive numeric value
ItemName	string?, max length 200 Description of the requested material

MaterialOrderService – Overview

Namespace	Application.Services.Admin
Interface	IMaterialOrderService
Responsibilities	Business logic for material orders CRUD operations and totals calculation Code generation and year listing OTP send & validate plus signing
Dependencies	IUnitOfWork IWebHostEnvironment IHttpContextAccessor ISMSService ISMSForSendingOTPService

INDEX

Material Order Module – Overview
MaterialOrder Entity
MaterialOrderItem Entity
MaterialOrderService – Overview
MaterialOrderService – Core Methods
MaterialOrderController – Overview
Helper – CheckLoggedUserIfTrainerAndReturnData
Controller Actions – Main
Controller Actions – Details & Reporting
OTP & SMS Approval Workflow

MaterialOrderService – Core Methods

GetAllAsync	Returns all material orders including: SignatureUser , SignatureManager , Department Used for listing, print, and export
GetAllSuppliersAsync	Returns lightweight supplier list: Id , Arabic name , English name
GetNewCodeAsync	Generates next sequential MaterialOrderCode Reads last Id and code, parses numeric part, increments, formats as 4 digits (D4)
GetByIdAsync	Retrieves single order by Id with: Items , SignatureUser , SignatureManager , Department
GetAllYearsInDb	Returns distinct years from MaterialOrder.Date Used for year-based filtering or reports
AddAsync	Adds new MaterialOrder Iterates items, ensures Quantity/SinglePrice non-null Calculates row totals and OrderTotal Ensures unique code; generates new if collision Saves and returns new Id
UpdateAsync	Updates existing order Loads current record, deletes existing items Adds new items and recalculates OrderTotal Preserves UserId and updates scalar fields via UpdateValues Saves changes
DeleteAsync	Deletes order and its items Removes related MaterialOrderItem records, then header
CheckIsMonthRegisteredBefore	Checks if another record exists in same year/month (excluding current Id) Used to prevent duplicate monthly records when required
SendOtpAsync	Generates OTP via OTPHelper.SaveOtpAsync Sends OTP using ISMSForSendingOTPService Returns true on SMS success
ValidateOtpAsync	Validates OTP using OTPHelper.ValidateOtpAsync Loads MaterialOrder by Id Fetches latest signature for current user If role = Trainer → sets SignatureUserId If role = Manager → requires trainer signature first, then sets SignatureManagerId Saves and returns success / message

INDEX

Material Order Module – Overview
MaterialOrder Entity
MaterialOrderItem Entity
MaterialOrderService – Overview
MaterialOrderService – Core Methods
MaterialOrderController – Overview
Helper – CheckLoggedInUserIfTrainerAndReturnData
Controller Actions – Main
Controller Actions – Details & Reporting
OTP & SMS Approval Workflow

MaterialOrderController – Overview

Namespace	FougeraClub.Areas.Admin.Controllers
Area	Admin
Authorization	[AdminAuthorize]
Purpose	Handle UI endpoints for material orders: listing, filtering, CRUD, printing, Excel export, OTP approval
Dependencies	IMaterialOrderService ITrainerService IMapper IUnitOfWork IHubContext<NotificationHub> INotificationService IHttpContextAccessor userManager<ApplicationUser>

Helper – CheckLoggedInUserIfTrainerAndReturnData

Attributes	[IgnoreAction] [NoLogging]
Return Type	Task<(string userId, int trainerId, int departmentId)>
Logic	Reads logged-in username (email) from HttpContext.User Loads ApplicationUser and associated Trainer If trainer exists → returns user id, trainer id, department id Else → returns user id and zeros for trainer/department Used to filter data and pre-fill forms for trainers

INDEX

Material Order Module – Overview

MaterialOrder Entity

MaterialOrderItem Entity

MaterialOrderService – Overview

MaterialOrderService – Core Methods

MaterialOrderController – Overview

Helper –
CheckLoggedUserIfTrainerAndReturnData

Controller Actions – Main

Controller Actions – Details & Reporting

OTP & SMS Approval Workflow

Controller Actions – Main

Index	Attributes: [YesGet] Lists material orders with filters: User, Department, DateFrom, DateTo Trainer user sees only own orders Uses <code>PaginatedList<MaterialOrderVM></code> (page size 50) Ajax support: returns <code>_ListPartial</code> when requested via XMLHttpRequest
AddEdit (GET)	Creates or edits material order Generates new code via service for new orders If trainer logged, pre-fills trainer, user, department and flags <code>IsTrainerLogged</code> Loads department and user dropdowns
AddEdit (POST)	Attributes: [HttpPost], [ValidateAntiForgeryToken] Reapplies trainer context (if any) On validation failure: rebind dropdowns and keep code On success: maps to <code>MaterialOrder</code> and calls <code>AddAsync</code> or <code>UpdateAsync</code> Redirects to Index (create) or back to AddEdit (edit)
Delete	Attributes: [HttpPost] Deletes order via service and redirects to Index

Controller Actions – Details & Reporting

Details	Attributes: [IgnoreAction], [YesGet] Read-only view of single order Loads applicant name from <code>ApplicationUser</code> Loads <code>TrainerSignedName</code> and <code>ManagerUserName</code> from signature users
Print	Attributes: [IgnoreAction], [YesGet] Printer-friendly list of orders Same filters as Index, no pagination Uses <code>MaterialOrderVM</code> list
PrintDetails	Attributes: [IgnoreAction], [YesGet] Printable single order Can include suppliers data for display
createExcelReport_Download	Attributes: [IgnoreAction], [YesGet] Filters like Index Builds <code>ExcelDataDTO</code> list with Code, Date, Department, Applicant Supports AR / EN via <code>ResourceI</code> and Session language Calls <code>ExcelStaticReport.ExcelReportArEn_</code> and returns XLSX file

INDEX

Material Order Module – Overview
MaterialOrder Entity
MaterialOrderItem Entity
MaterialOrderService – Overview
MaterialOrderService – Core Methods
MaterialOrderController – Overview
Helper – CheckLoggedUserIfTrainerAndReturnData
Controller Actions – Main
Controller Actions – Details & Reporting
OTP & SMS Approval Workflow

OTP & SMS Approval Workflow

SendOtp	Attributes: [IgnoreAction], [NoLogging], [HttpPost] Calls service SendOtpAsync Returns JSON: { success = true/false }
ValidateOtp	Attributes: [HttpPost], [IgnoreAction] Accepts OtpValidationRequest (Id, Code, Role) Validates OTP via service If Role = Trainer and success → sends SignalR and notification to Manager role about new request ready for approval Returns JSON with success and message
Trainer Role	Signs as applicant (SignatureUserId) Triggers notifications to managers
Manager Role	Signs as manager (SignatureManagerId) Requires Trainer signature to exist first